

Symbiosis Institute of Technology, Pune

Faculty of Engineering

CSE- Academic Year 2026-27

Java Programming – Lab Batch 2024-28

Lab Assignment No :- 9	
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Batch	2025-2026
Class	CSE-A1
Academic Year & Semester	2026 - 3rd Semester
Date of Submission	10th-july-2026
Title of Assignment:	Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.

Practice questions
 	
Output Screenshots
Solve questions on given link and attach screen shot of leetcode with your login and o/p
	PART-1
	`/*`
	` * Assignment No: 9`
	` * Program 1: Bank Account using final Keyword`
	` *`
	` * Aim:`
	` * To demonstrate the use of the final keyword`
	` * with variables in Java.`
	` */`

```
// BankAccount class
class BankAccount {

    // final variable (cannot be changed after initialization)
    final int accountNumber;

    // Other data members
    String accountHolderName;
    double balance;

    // Parameterized constructor
    BankAccount(int accountNumber, String accountHolderName, double balance) {

        // Initialize final variable
        this.accountNumber = accountNumber;

        // Initialize remaining variables
        this.accountHolderName = accountHolderName;
        this.balance = balance;
    }

    // Method to display account details
    void displayAccountDetails() {

        System.out.println("------ Bank Account Details ------ ");
        System.out.println("Account Number : " + accountNumber);
        System.out.println("Account Holder : " + accountHolderName);
        System.out.println("Balance        : ₹" + balance);
    }

    /*
     * Uncommenting the below method will produce
     * a compilation error because accountNumber
     * is declared as final.
     */
```

```
    /*
    void changeAccountNumber() {
        accountNumber = 12345; // ERROR
    }
    */
}

// Main class
public class BankAccountDemo {

    public static void main(String[] args) {

        // Create BankAccount object
        BankAccount account = new BankAccount(
                100101,
                "Abhishek Singh",
                50000
        );

        // Display account details
        account.displayAccountDetails();
    }
}
```

PART 2

```
/*
 * Assignment No: 9
 * Program 2: Library Book Management using final
Keyword
 *
 * Aim:
 * To demonstrate the use of final keyword
 * with variables.
 */

// LibraryBook class
class LibraryBook {

    // final variable
    final String ISBN;
```

```
    // Other book details
    String title;
    String author;
    double price;

    // Parameterized constructor
    LibraryBook(String ISBN, String title, String
author, double price) {

        // Initialize final variable
        this.ISBN = ISBN;

        // Initialize other variables
        this.title = title;
        this.author = author;
        this.price = price;
    }

    // Display book details
    void displayBookDetails() {

        System.out.println("------ Library Book
Details ------ ");
        System.out.println("ISBN   : " + ISBN);
        System.out.println("Title  : " + title);
        System.out.println("Author : " + author);
        System.out.println("Price  : ₹" + price);
    }

    /*
     * The following code is invalid because
     * ISBN is declared final.
     */

    /*
    void changeISBN() {
        ISBN = "9780000000"; // ERROR
    }
    */
}

// Main class
```

```java
public class LibraryBookDemo {

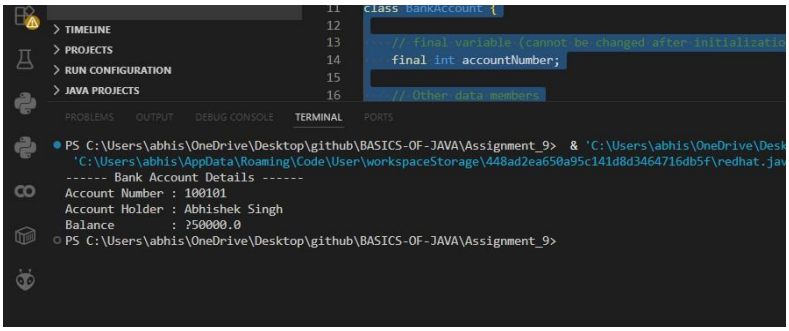
    public static void main(String[] args) {

        // Create LibraryBook object
        LibraryBook book = new LibraryBook(
                "978-81-123456-7-8",
                "Programming with Java",
                "James Gosling",
                699.00
        );

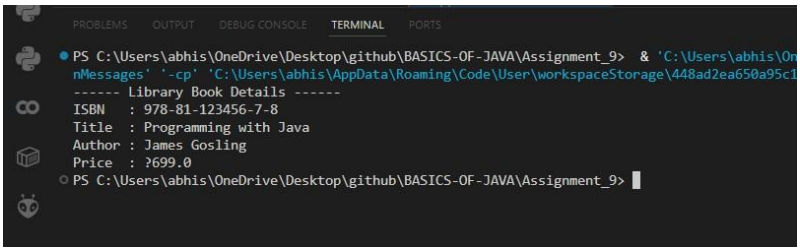
        // Display book information
        book.displayBookDetails();
    }
}
```

Output Screenshots (if applicable)

Part-1



Part-2



Conclusion	This assignment demonstrated the use of the **`final` keyword** in Java with variables. The `final` variables (`accountNumber` and `ISBN`) were initialized through constructors and could not be modified afterward, ensuring data integrity and immutability. Attempting to change these values results in a compile-time error. Through the **Bank Account** and **Library Book Management** programs, the assignment highlighted how the `final` keyword helps protect important data and improves the reliability and security of object-oriented programs.